VECTORS

SCALAR QUANTITIES - have magnitude or size only.

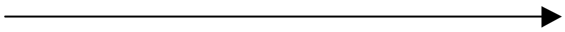
e.g. mass, distance, speed, density, temperature.

VECTOR QUANTITIES - have magnitude and direction.

e.g. displacement, velocity, acceleration, force, impulse, momentum.

Vectors are represented by arrows - the length indicates the magnitude of the vector and the arrowhead indicates the direction in which it acts.

i.e.



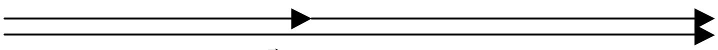
COMBINING VECTORS

ADDITION

Vectors are added head-to-tail. The resultant vector runs from the tail of the first vector to the head of the last vector.

1. **ADDITION**

ONE-DIMENSION

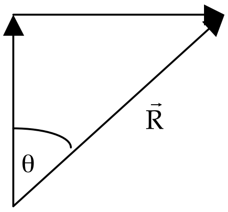


The resultant vector R = 25 .0 N - it runs from the tail of the first vector to the head of the last vector.

TWO DIMENSIONS

VECTORS AT RIGHT ANGLES

Use Pythagorus' Theorem and trigonometry to find the resultant.



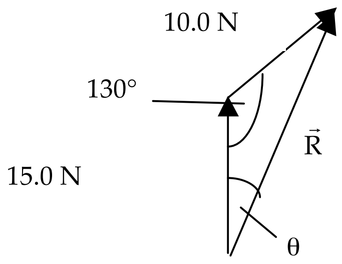
$$\vec{R} = \sqrt{(10.0)^2 + (15.0)^2} = 18.03 \text{ N}$$

$$\tan\theta = \frac{10.0}{15.0} \Rightarrow \theta = 33.69°$$

$\therefore$ <u>$\vec{R}$ = 18.0 N at 33.7° to the 15.0 N vector</u>.

VECTORS NOT AT RIGHT ANGLES

Use the sine and cosine rules for triangles to find the resultant vector.



$$\vec{R} = \sqrt{(15.0)^2 + (10.0)^2 - 2(15.0)(10.0)\cos 130°}$$

$$= 22.76 \text{ N}$$

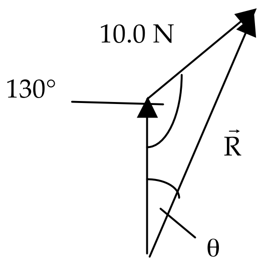
$$\frac{10.0}{\sin\theta} = \frac{22.76}{\sin 130°}$$

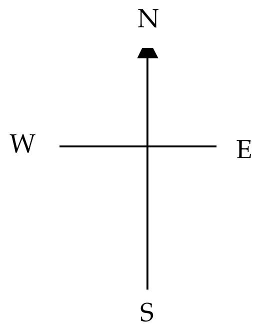
$$\Rightarrow \theta = 19.67°$$

$\therefore$ $\underline{\vec{R} = 22.8 \text{ N at } 19.7° \text{ to the } 15.0 \text{ N vector}}$.

NOTE
If compass directions are given, it makes it easier to describe the direction.

Consider the example above.





The resultant vector R is at a 19.7° angle east of north.

$\therefore$ Direction is N 19.7° E.

Also, north can be considered the 0° bearing, with all other bearings determined clockwise from this direction.

$\therefore$ Direction is a bearing of 19.7°.

<u>Hence R = 22.8 N N 19.7° E or R = 22.8 N on a bearing of 19.7°</u>.

2. **SUBTRACTION**

ONE-DIMENSION

The easiest way to deal with vectors in one dimension is to **choose one direction as positive**.

Any vector in this direction is positive while all those in the other direction are negative.

It is then a simple matter of adding numbers.

EXAMPLE 1
A car moving at 15.0 ms^{-1} brakes suddenly to avoid a dog and reduces to a velocity of 5.00 ms^{-1}. Calculate its change in velocity.

Take the forwards direction as positive.

By definition the change in velocity is given by :

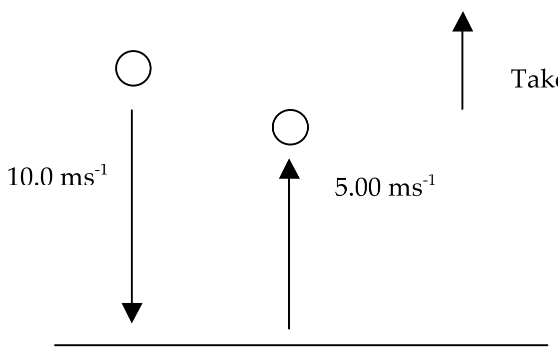
$$\Delta v = v - u$$
$$= 5.00\ ms^{-1} - 10.0\ ms^{-1}$$
$$= -5.00\ ms^{-1}$$

$\therefore$ <u>Change in velocity = 5.00 ms^{-1} backwards.</u>

The answer is negative which means the resultant is in the opposite direction to the one you chose as positive.

i.e. The direction is backwards.

EXAMPLE 2
A ball falls onto a concrete floor at 10.0 ms^{-1} and rebounds at 5.00 ms^{-1}. Calculate the change in velocity of the ball.



Take up as positive.

u = -10.0 ms^{-1} since it is down in the negative direction.

$$\Delta v = v - u$$
$$= 5.00 - (-10.0)$$
$$= \underline{15.0\ ms^{-1}\ \text{upwards}}$$

Answer is positive so the final direction is upwards in the positive direction chosen.

2. TWO DIMENSIONS

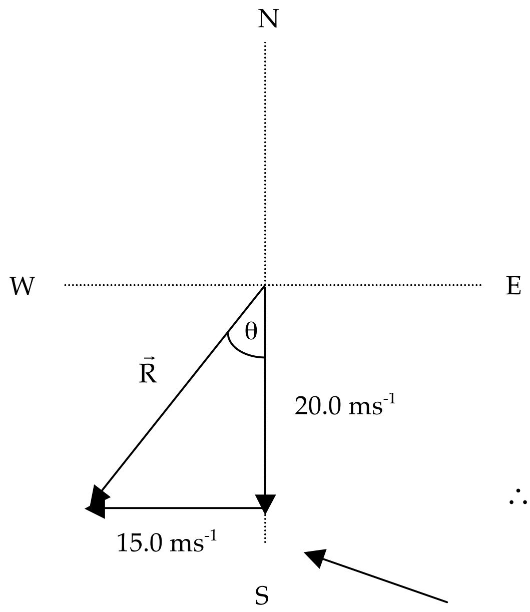
When subtracting vectors, the "negative" sign means:

"change the direction of the subtracted vector 180° and add it to the other vector".

This works best when vectors act at an angle to each other.

EXAMPLE

A car moving at 20.0 ms^{-1} north makes a left hand turn around a sweeping bend and heads west at 15.0 ms^{-1}. Calculate its change in velocity.



$$\Delta v = v - u$$

$$= 15.0\ ms^{-1}\ W - 20.0\ ms^{-1}\ N$$

$$= 15.0\ ms^{-1}\ W + 20.0\ ms^{-1}\ S$$

$$\vec{R} = \sqrt{(15.0)^2 + (20.0)^2} \qquad \tan\theta = \frac{15.0}{20.0}$$

$$= 25.0\ ms^{-1} \qquad \Rightarrow \quad \theta = 36.87°$$

$$\therefore \quad \underline{\Delta v = 25.0\ ms^{-1}\ S\ 36.9°W}$$

The use of a compass grid line makes it easier to determine the direction of the resultant vector.